

Amendment 'A Safe Bet' Ethics Application

OVERVIEW OF AMMENDEMENT:

Amendment 1: Changes to Quantitative Data Collection Period

- Extended the original 6-month data collection period to 10 months.
- Introduced three phases: 1) POC Preparation Data Collection (1 month), 2) POC Data Collection (6 months), and 3) Extended Data Collection for BETA Preparation (3 months).

Amendment 2: Changes to the Participant Inclusion Threshold

- Increased the minimum Active Gambling Days (AGD) from 15 to 20 for participant eligibility.

Amendment 3: Changes in Participant Consent Procedures (UPDATED: following comments by Tijs Gelens)

- Streamlined consent procedures in collaboration with privacy officers.
- Moved away from explicit informed consent to an approach aligned with operators' existing privacy policies.

Amendment 4: Changes to Sample Size and Recruitment Procedures

- Revised the initial sample size from 200,000 to ~8,000 per operator using a stratified sampling method.
- Introduced a rolling recruitment strategy to replenish the sample based on stratified risk categories.

Each of these amendments aims to address challenges in the project while aligning with ethical standards and practical constraints. They collectively serve to strengthen the study's methodology, ethical considerations, and overall feasibility.

AMMENDMENTS

1. Changes to Quantitative Data Collection Period

Under the original application, we stated that the quantitative arm of the study would last a period of 6 months. However, 6 months only accounts for the active data collection period with the live POC RG framework. As such, we provided an amended overview to encompass the phased timeline.

1. **POC Preparation Data Collection:** One month prior to the launch of the POC RG framework live on the gambling interface, we will collect begin to collect player behavioral tracking data. This is required to accurately assess and provide risk-level assignments for tailored RG tool recommendations at the study period onset. This month is for baseline data collection only and extends the study period by one month before the POC commences.
2. **POC Data Collection Period:** Immediately following the baseline data collection, the RG framework will go live and run for a period of 6 months as previously specified. During this time, we will collect data under the same aims; focusing on feasibility, user acceptance, and the potential impact of the RG framework.
3. **Extended Data Collection for BETA Preparation:** After the conclusion of the POC study period, the RG framework will remain active for an additional 3 months on

gambler accounts. Though the data from these three months won't be utilised under the current application, when combined with the POC data set it will serve as a 9-month baseline period required for the full-scale BETA impact evaluation.

4. **Total Duration:** The data collection period is 10 months; 1 month for preparation data collection for the POC risk assignment, with the RG framework active for a total duration of 9 months: 6 months for the POC, and an additional 3 months for extended data collection aimed at preparing for the BETA phase.

NB. This amendment aligns with our initial application, where it was indicated that this was Phase 1 and a subsequent Phase 2 was to follow. We are now in the process of preparing the ethics application for the BETA phase. However, we felt that an amendment of the current application was necessary for ethical considerations and to provide a foundational structure for the upcoming BETA impact evaluation.

2. Changes to the Participant Inclusion Threshold

In our initial ethics application, we stated that participants must have at least 15 active gambling days (AGD) in the prior period for accurate risk assessment. Due to operational constraints and compliance requirements, this has been **updated to a minimum of 20 AGD**. This adjustment not only aligns with the technical constraints of the risk assessment algorithm but also strengthens the safety measures, ensuring a more reliable and compliant risk-level assignment for participants. This change ensures both our research and the operators adhere to the highest standards of safety and compliance.

3. Changes in Participant Consent Procedures (See below for clarifications)

After consultation with the privacy officers of the collaborating gambling operators in both the United Kingdom and the Netherlands and in coordination with the EUR Data Privacy Officer, we ask to amend our participant consent procedures for the quantitative arm of the study.

Rationale for Amendment:

1. **Operator Privacy Policies (informing players):** EUR is not collecting the data from the players, the operators do. Hence, both the UK and NL operators' privacy statement explains clearly that the operators need to collect data to comply with their legal obligations (UK and Dutch online gambling regulations). Additionally, the privacy statements include provisions for sharing anonymised and/or personal data with third parties for research and development purposes. As a result, players have been informed about the data collection and sharing thereof and agreed to these conditions by signing up to and using the operator's services. Further information will be provided to the players with the release of the new interface (introduced by both operators).
2. **Receiving anonymous data:** All data transferred to the EUR is extensively anonymised (for the EUR) by the operators. The operators will not share details through which the EUR can identify the individual players. This adheres to privacy regulations and safeguards participant confidentiality.
3. **Data Privacy Officer Collaboration:** The data collection and transfer procedures, including this amendment, have been drafted in collaboration with the EUR Data Privacy Officer, ensuring full compliance with relevant privacy laws.

Impact of Changes:

1. **Informed Consent:** Given that players have already been informed and agreed to collection and sharing of their data via the operators' privacy statements and that all transferred data will be extensively anonymised, we have determined that obtaining explicit informed consent (i.e., the previous "opt-out" option) is not necessary for the quantitative arm of this research. By registering on the operator's website and agreeing to operator's privacy policy and terms of use, players already consent to participate in the evaluation and monitoring of responsible gambling measures.
2. **Notification Changes:** The operators will notify the players about the release of the new interface, their updated responsible gambling policy, and the data sharing. The notification will be in the form of a pop-up message, which all players will see upon their next login to the operators' site. They will not have an option to opt out of this specific research initiative.
3. **Removal of Exit:** The pop-up notification will not include an "exit" icon but will include an "OKAY" button asking for players confirmation that they have read and understood the notification message. This ensures that players acknowledge the information to continue using the platform, in line with the operators' existing terms and privacy policies.
4. **Data Privacy:** The responsibility for data use and privacy conditions continues to lie with the operators up until the sharing of the data. Once data is shared with EUR, the EUR is responsible for data use and privacy conditions.
5. **Transparency and Accessibility of Information:** Besides the pop-up notification, information on the research aims will be available to players anytime through the dedicated, responsible gambling homepage on both UK and NL sites.

NB. In support of the amendments to participant consent procedures, we have uploaded two documents: 1) An updated version of the player notification when gamblers receive the POC RG interface; and 2) a data flow document crafted by (Bo Schijven), EUR Data Privacy Officer, providing a comprehensive and visual representation of the data flow specific to the POC phase of the study.

Clarifications Following Review of Tijs Gelens:

- **Existing Data Collection by Operators:** It's important to emphasise that the data we are utilising will be collected by the gambling operators regardless of the researchers involvement. We are not involved in the primary data collection process. Instead, we are working with secondary data, which means that we do not directly interact or intervene with the participants.
- **Prior Consent Given by Gamblers:** Upon registration to the gambling platforms, gamblers have provided their explicit consent to be part of scientific research as part and have consented for the operators to share their data with third parties, including research institutions for scientific research purposes. This includes consenting to the monitoring and evaluation of RG tools, a cornerstone of our study. The general terms and conditions of both operators have been reviewed by the EUR privacy officer to verify these consent measures are taken and maintained by the gambling operators. We effectively become data controllers only when these anonymized datasets are shared with us by the operators.

- **Mandatory Obligations & The KOA Law:** The KOA law regulates online gambling activities, including the mandatory obligation of operators to evaluate and research responsible gambling and addiction prevention measures. Any individual wishing to gamble on such a platform must provide consent to these activities, as stipulated by the law. This is an imperative point because their consent is provided upstream, and this law supersedes the Dutch scientific and ethics code of conduct. Given the prevailing legal framework, we cannot and should not seek an additional layer of consent, as this would countermand the existing legal structure.
- **Notification and Transparency:** Players who are part of the POC study (i.e., included within the data collection pool) are informed about the specifics, objectives, and implications of the study. With the release of a POC new interface, all participants will receive explicit and transparent information about the research's scope and what it involves.
- **Protection of Participant Data:** All data that is shared with EUR is anonymized by the operators before transfer. We, as researchers, do not have the capability or means to identify individual participants from the data we receive. This ensures that the participants' privacy is upheld and there is a clear demarcation between the operator's data and what is shared with us.

4. Changes to Sample Size and Recruitment Procedures

Our initial ethics application stipulated a desired sample size of 200,000 players, assuming a full-scale launch across both operators. In a scenario where this wasn't feasible, a random sample of 30,000 players was intended as an alternative.

Due to constraints and further insights into the operator customer base, we've revised the sampling strategy for the POC phase. In the initial phase of the POC we're employing a stratified sampling method that includes about ~8,000 players per operator. These players are sampled based on risk-level proportions: high-risk, 10% of the sample (approximately 800 players); moderate-risk, constituting 35% (around 2,800 players); and low-risk, making up the remaining and largest proportion at 55% (about 4,400 players). All gamblers in this initial sample must meet the criterion of 20 Active Gambling Days (20AGD) on the first day of POC data collection. The goal is to ensure this sample is representative of the larger player base, which includes 100,000-120,000 gamblers per operator. We have intentionally oversampled high-risk and medium-risk players to gather more data from these important segments. This stratified approach aims to create a balanced and representative initial dataset for evaluating RG framework across different risk profiles.

In addition to our initial stratified sample, we're amending the initial sampling method to include a rolling recruitment strategy as the POC progresses. As players “drop out”—defined as those accumulating fewer than 20 Active Gambling Days within a 90-day evaluation period—we will replenish the sample using the same stratified risk categories: 10% high-risk, 35% medium-risk, and 55% low-risk. Both the initial sample and the newly recruited players will be tagged at entry for identification and future analysis. Gamblers defined as “drop-outs” are not excluded from the POC study but are tagged as such. Their re-engagement with the gambling platform will be monitored, and their data will be retained if they meet the study criteria at a later stage. Additionally, we note that new participants entering the

study receive the same notifications and are fully informed of the study objectives, their rights, and how their data will be used.

Rational for Rolling Recruitment

Our rolling recruitment strategy serves several essential functions for the integrity and success of our study. First, one of the critical requirements for the long-term success of this project is the accumulation of a sufficiently large and diverse dataset needed for the development of our machine-learning-based recommendation system, a crucial feature of the BETA phase. A fixed-sample approach, while providing valuable initial insights, is not adequate for this extended data requirement due to the natural attrition and fluctuation in player behavior.

Second, the rolling recruitment can improve the ecological validity of our RG tool set evaluations. By replenishing and expanding our sample systematically, we are able to capture a broader range of player behaviours, and responses to RG tools. Moreover, it also ensures the randomiser at the POC level can make as many diverse RG tool recommendations as possible, facilitating more comprehensive testing and evaluation of these tools for the recommendation system.

Third, the need for rolling recruitment is further underscored by the naturalistic setting of our study, which necessitates the ability to adapt to fluctuating risk levels, real-world behavioural changes, and industry-specific challenges such as player churn. Although we adjusted our initial sample size to account for expected attrition, the operators have provided statistics on the average monthly churn rate, which suggests a cumulative impact on the sample over time. As a result, we project the final sample size for the full 9-month POC phase to reach approximately X.

Finally, the player tagging system on entry adds a layer of methodological rigor to our study. By tagging new players upon entry, we are able to maintain the integrity of the initial sample. This means we are able to analyse attrition rates separately for the initial and rolling samples, thereby providing valuable insights into whether the study design or other factors are affecting engagement. Most importantly, the tagged groups ensure that the statistical power of the study remains robust. By knowing which cohort each player belongs to, we can apply more precise statistical models that account for potential confounders, thereby strengthening the validity and reliability of our findings. This is particularly crucial for the BETA stages, where understanding nuanced player behaviour will be essential for the machine learning models' training and validation.

Connection to Previous Amendments:

- **Data Collection (Amendment 1):** The extension of the data collection period such that the RG framework will be live for 9 months, inclusive of the POC and the BETA, The rolling recruitment strategy becomes necessary to ensure the data volume required for the BETA baseline dataset.
- **Participant Inclusion Threshold (Amendment 2):** The adjusted requirement for 20 Active Gambling Days (20AGD) serves as the benchmark for not only initial recruitment but also for the continuous enrollment of participants throughout the POC phase.

- **Consent Procedures (Amendment 3):** Any new players who join during the POC will be notified as per the amended notification procedures, ensuring the onboarding of new recruits is aligned to data privacy and regulator requirements.